

Statistics Formula Sheet with Examples (Bangla + English)

Mean (গড়)

$$\bar{x} = \Sigma x / n$$

Example: 10,20,30 → Mean=(60/3)=20

Median (মাঝারি)

Example: 2,4,6,8,10 → Median=6

Mode (মোড)

Example: 1,2,2,3 → Mode=2

Range

Example: Max=50, Min=10 → Range=40

Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Example: {1,2,3} → Variance≈0.67

Standard Deviation

Example: Variance=9 → SD=3

Probability

$$P(E) = \text{Favourable} / \text{Total}$$

Example: Dice → P(3)=1/6

Conditional Probability

$$P(A|B) = P(A \cap B) / P(B)$$

Example: 0.30/0.60=0.50

Bayes Theorem

Example: P(B|A)=0.8, P(A)=0.2, P(B)=0.4 → P(A|B)=0.40

Permutation

$${}_5P_2 = 20$$

Combination

$${}_5C_2 = 10$$

Z-Score

$$z = (x - \mu) / \sigma$$

$$\text{Example: } (70 - 50) / 10 = 2$$

Correlation

Example: $r=0.9 \rightarrow$ strong positive relation

Quartile

Example: Data 1-9 $\rightarrow Q_2=5$

Percentile

Example: $P_{50} = \text{Median}$

IQR

Example: $Q_3=30, Q_1=10 \rightarrow \text{IQR}=20$